Alert Reader Study: alerts and options

20 real alerts from the trained ensemble, in a fixed mixed order. For each alert the participant answers four questions. The answer key is on the last pages; do not show it to participants.

Alert 1 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 100%, against 20%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       20% of the total evidence, strongest at record 7 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       9% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 6 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.05s. 'Attack' is the positive class.)
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1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average gap between replies from the device (Bwd_IAT_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 2 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 92%, against 18%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       18% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       12% of the total evidence, strongest at record 9 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 8 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 9 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.85s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 3 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 98%, against 19%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 98% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       23% of the total evidence, strongest at record 9 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       7% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       14% of the total evidence, strongest at record 9 of 10.
  3. the average packet size in the conversation (Pkt_Len_Mean)
       8% of the total evidence, strongest at record 10 of 10.
  4. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       7% of the total evidence, strongest at record 10 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all
detectors were cnn 34%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.98s. 'Attack' is the positive class.)
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1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average gap between packets in the conversation (Flow_IAT_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 4 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 98%, against 20%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       19% of the total evidence, strongest at record 7 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       14% of the total evidence, strongest at record 9 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 9 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 7 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.99s. 'Attack' is the positive class.)
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1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 5 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.0%
at k=2, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       13% of the total evidence, strongest at record 8 of 10.
  2. how much data the device said it was ready to receive when it first replied
(Init_Bwd_Win_Byts)
       11% of the total evidence, strongest at record 6 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       9% of the total evidence, strongest at record 8 of 10.
  5. the total size of the addressing information on the device's replies (Bwd_Header_Len)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       11% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.81s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 6 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 100%, against 20%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       22% of the total evidence, strongest at record 8 of 10.
  4. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       10% of the total evidence, strongest at record 9 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 4 of 10.

Detector agreement: the bilstm detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.87s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 7 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.3%
at k=3, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
       9% of the total evidence, strongest at record 8 of 10.
  3. the longest pause between packets sent to the device (Fwd_IAT_Max)
       8% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average gap between replies from the device (Bwd_IAT_Mean)
       8% of the total evidence, strongest at record 1 of 10.
  4. the shortest gap between replies from the device (Bwd_IAT_Min)
       7% of the total evidence, strongest at record 1 of 10.
  5. the average gap between packets in the conversation (Flow_IAT_Mean)
       7% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.85s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average gap between replies from the device (Bwd_IAT_Mean)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ the average gap between packets in the conversation (Flow_IAT_Mean)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the longest pause between packets sent to the device (Fwd_IAT_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 8 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 20%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       22% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 7 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 7 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 7 of 10.
  5. the size of the largest packet in the conversation (Pkt_Len_Max)
       6% of the total evidence, strongest at record 7 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all
detectors were cnn 34%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.74s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 9 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -0.0%
at k=2, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       10% of the total evidence, strongest at record 9 of 10.
  4. the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.
  5. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       5% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       18% of the total evidence, strongest at record 1 of 10.
  3. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       8% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.98s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 10 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 19%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       24% of the total evidence, strongest at record 9 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       12% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       7% of the total evidence, strongest at record 2 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all
detectors were cnn 34%, bilstm 33%, transformer 34%.
(Explanation produced by SHAP in 1.01s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average quiet gap between bursts of activity (Idle_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 11 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.1%
at k=3, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 4 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 7 of 10.
  5. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       11% of the total evidence, strongest at record 2 of 10.
  3. the average size of packets in the conversation (Pkt_Size_Avg)
       8% of the total evidence, strongest at record 5 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.00s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 12 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -0.0%
at k=2, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. how much data the device said it was ready to receive when it first replied
(Init_Bwd_Win_Byts)
       13% of the total evidence, strongest at record 4 of 10.
  3. the total size of the addressing information on the device's replies (Bwd_Header_Len)
       7% of the total evidence, strongest at record 4 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       7% of the total evidence, strongest at record 1 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 4 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 6 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.86s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 13 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +1.0%
at k=5, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       13% of the total evidence, strongest at record 9 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       9% of the total evidence, strongest at record 10 of 10.
  5. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       7% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       10% of the total evidence, strongest at record 10 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       9% of the total evidence, strongest at record 10 of 10.

Detector agreement: the bilstm detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.03s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average quiet gap between bursts of activity (Idle_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 14 of 20

```
[VERIFIED] Removing the top 2 cited measurements changes this decision by 14%, against 0%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       9% of the total evidence, strongest at record 3 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       9% of the total evidence, strongest at record 9 of 10.
  5. the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       14% of the total evidence, strongest at record 2 of 10.
  4. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       7% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.96s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 15 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -13.4%
at k=5, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 68% confidence.

What pointed to Attack:
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 6 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       5% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       18% of the total evidence, strongest at record 2 of 10.
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       11% of the total evidence, strongest at record 9 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       6% of the total evidence, strongest at record 1 of 10.

Detector agreement: the bilstm detector carried the most weight (39%); weights across all
detectors were cnn 32%, bilstm 39%, transformer 29%.
(Explanation produced by SHAP in 8.19s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how long the connection lasted (Flow_Duration)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 16 of 20

```
[VERIFIED] Removing the top 5 cited measurements changes this decision by 9%, against 0% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       12% of the total evidence, strongest at record 8 of 10.
  4. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       4% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       13% of the total evidence, strongest at record 2 of 10.
  3. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 8 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       4% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.93s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ how long the connection lasted (Flow_Duration)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 17 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +1.7%
at k=3, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       8% of the total evidence, strongest at record 5 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       5% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       19% of the total evidence, strongest at record 9 of 10.
  4. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       5% of the total evidence, strongest at record 9 of 10.
  5. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.08s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 18 of 20

```
[VERIFIED] Removing the top 2 cited measurements changes this decision by 7%, against 0% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       7% of the total evidence, strongest at record 8 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       7% of the total evidence, strongest at record 9 of 10.
  5. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       5% of the total evidence, strongest at record 10 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       19% of the total evidence, strongest at record 9 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       6% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.95s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the longest quiet period in the conversation (Idle_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 19 of 20

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.0%
at k=5, threshold 5%) — the model's decision here is spread across many measurements, so no
short list explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. how long the conversation lasted (Flow_Duration)
       12% of the total evidence, strongest at record 10 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       8% of the total evidence, strongest at record 8 of 10.
  4. the longest quiet period in the conversation (Idle_Max)
       7% of the total evidence, strongest at record 10 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       10% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.05s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how long the conversation lasted (Flow_Duration)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Alert 20 of 20

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 19%
for unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       18% of the total evidence, strongest at record 7 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       5% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       9% of the total evidence, strongest at record 10 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port
scans and SYN attacks look) (SYN_Flag_Cnt)
       6% of the total evidence, strongest at record 2 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.

Detector agreement: the bilstm detector carried the most weight (34%); weights across all
detectors were cnn 33%, bilstm 34%, transformer 33%.
(Explanation produced by SHAP in 0.93s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Answer key (facilitators only)

Scoring rule for question 1: correct if the reader ticks at least one measurement that pointed to Attack, and ticks no counter-evidence and no distractor.

#	Type	Correct (pointed to Attack)	Wrong: counter-evidence	Wrong: distractors
1	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Fwd_IAT_Mean, Bwd_IAT_Mean
2	Certified	Pkt_Size_Avg	SYN_Flag_Cnt, Bwd_Seg_Size_Avg, Pkt_Len_Max, ACK_Flag_Cnt	Bwd_IAT_Min, Idle_Max
3	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, Pkt_Len_Mean, SYN_Flag_Cnt	Flow_IAT_Mean, Fwd_IAT_Mean
4	Certified	Pkt_Size_Avg	SYN_Flag_Cnt, Bwd_Seg_Size_Avg, Pkt_Len_Max, ACK_Flag_Cnt	Fwd_IAT_Tot, Init_Bwd_Win_Byts
5	Uncertified	Fwd_Pkt_Len_Min, Init_Bwd_Win_Byts, Protocol, Bwd_Header_Len	SYN_Flag_Cnt	ACK_Flag_Cnt, Pkt_Len_Max
6	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	ACK_Flag_Cnt, Idle_Max
7	Uncertified	Fwd_IAT_Tot, Fwd_IAT_Max	Bwd_IAT_Mean, Bwd_IAT_Min, Flow_IAT_Mean	Fwd_Pkt_Len_Max, Pkt_Size_Avg
8	Certified	Pkt_Size_Avg	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean, Pkt_Len_Max	Fwd_Pkt_Len_Mean, Fwd_IAT_Tot
9	Contradictory	Fwd_Pkt_Len_Max, Fwd_Pkt_Len_Mean, Bwd_Seg_Size_Avg	SYN_Flag_Cnt, Fwd_Pkt_Len_Min	Bwd_Header_Len, Fwd_IAT_Mean
10	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Idle_Mean, Bwd_Header_Len
11	Uncertified	Bwd_Seg_Size_Avg, Pkt_Len_Mean, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Pkt_Size_Avg	Fwd_Pkt_Len_Min, Pkt_Len_Max
12	Uncertified	Init_Bwd_Win_Byts, Bwd_Header_Len, Protocol	SYN_Flag_Cnt, ACK_Flag_Cnt	Pkt_Size_Avg, Pkt_Len_Max
13	Uncertified	Bwd_Seg_Size_Avg, Pkt_Len_Mean, Fwd_Pkt_Len_Max	Pkt_Size_Avg, SYN_Flag_Cnt	Fwd_IAT_Mean, Idle_Mean
14	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max, Fwd_Pkt_Len_Mean	SYN_Flag_Cnt, Fwd_Pkt_Len_Min	Pkt_Len_Min, Fwd_IAT_Tot
15	Contradictory	Bwd_Seg_Size_Avg, Pkt_Len_Mean	SYN_Flag_Cnt, Pkt_Size_Avg, Protocol	Flow_Duration, Fwd_IAT_Tot
16	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Pkt_Len_Max, ACK_Flag_Cnt	Bwd_IAT_Min, Flow_Duration
17	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Fwd_Pkt_Len_Min, Pkt_Len_Max	Fwd_Pkt_Len_Mean, Pkt_Len_Min
18	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max, Bwd_Seg_Size_Avg	SYN_Flag_Cnt, Pkt_Len_Max	Fwd_IAT_Mean, Idle_Max

19	Uncertified	Flow_Duration, Bwd_Seg_Size_Avg, Idle_Max, Pkt_Len_Mean	Pkt_Size_Avg	Fwd_Pkt_Len_Mean, Fwd_IAT_Mean
20	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Fwd_IAT_Tot, Idle_Max